Exp: No: 4

Date: 5/8/26 4: Setting up a Local Area Network (LAN)

1. AIM:

To design, setup, and configure a local Area Network (LAN) Connecting four computers through a network switch, Assign static ip address to all devices, verify connectivity using the ping command and demonstrate resource / data sharing across the network.

2. Equipment Required

1.	personal computer (PC)	quality (4)	Ethernet (NIC) enabled
2.	Network switch.	1	8/16/24 port unmanaged / managed.
3.	Ethernet cables.	4	straight - through. RJ 45 Connect

3. Theory:

A local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as home, office, or laboratory, allowing them to share resources such as files, printers, and internet connections.

4. Network Diagram.

The LAN was implemented using star topology as shown below. 4 pc connected to the control switch.

5. Ip address configuration table.

Device	Ip address	Subnet Mask	Connected to.
PC0	10.1.1.1	255.0.0.0	switch port Fa0/1
PC1	10.1.1.2	255.0.0.0	switch port Fa0/2.
PC 2	10.1.1.3	255.0.0.0	switch port Fa0/3.
PC 3	10.1.1.4	255.0.0.0	switch port Fa0/4.

6. Implementation in cisco packet tracer.

Step1: place device.

1. From device catogory panel - select end devices. and drag 4 PC onto work space.
2. select switches drag 1 switch onto workspace.

Step2: cable devices.

1. click connection - select copper straight - through cable. give connection to all PC's.

Step3: Assign Ip address to PC's:

1. click a PC → Desktop → IP configuration.

Enter the static Ip address and Subnet Mask as per the IP table.

Step 5: Verify connectivity using ping.

1. Click a PC → Desktop → command prompt.
2. using ping command. to test connectivity to the other pc's eg: 10.1.1.2.
3. successful "reply from" message confirm connectivity.

Step 6: Demonstrate Data sharing.

1. add a Server device and connect it to the switch.
2. Assign it an Ip address. via its desktop → IP config tab.
3. open the Servers services (config) tab and enable the HTTP service.
4. From another PC, open the web browser and enter the Server's Ip address to load the page, confirming successful data transfer across lan.

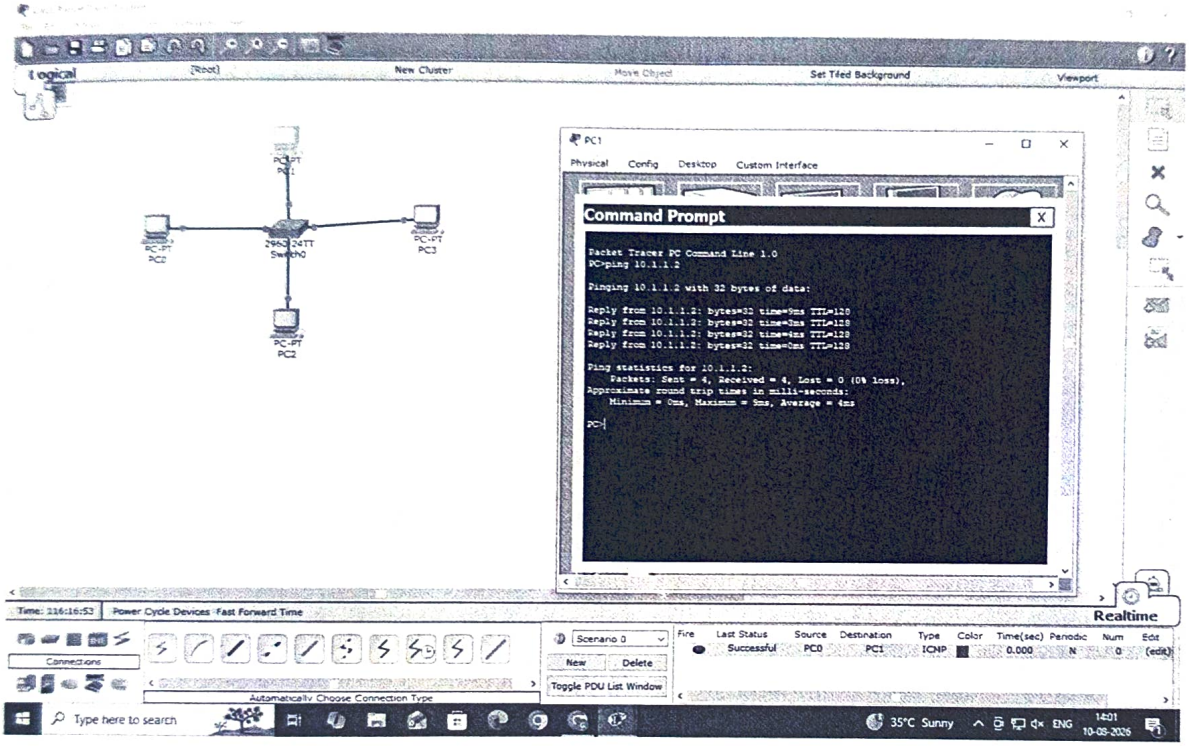
Step 7: observe the Packet flow in simulation mode:

1. Switch to Simulation mode.

send a ping or web request between two devices. and use the play / capture controls to step through the packet.

7. Observation.

All four PC's were successfully connected to the switch. in a star topology.



Result:

Thus the setting up a local area network (LAN) was implemented and executed successfully.